

Exercise (Very similar to a question in HOTS): I flip two coins hidden from your view.

Version One: You ask, "Is at least one of the coins heads?" I respond, "Yes."

What is the probability they are both heads (from your perspective)?

Version Two: I say "At least one is heads." What is the probability they are both heads (from your perspective)?

Version Three: You happen to see one of the coins, and notice that it is heads.

What is the probability they are both heads (from your perspective)?

Version One:

$A = \{\text{at least one heads}\}$

$B = \{\text{both heads}\}$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(B)}{P(A)}$$

$$= \frac{1/4}{1 - 1/4} = 1/3$$

HH HT
TH ~~TT~~

Version Two: Here I tell you "at least one heads". How did I decide what information to provide?

A reasonable model for the info. I provide

<u>Result</u>	<u>Info.</u>
HH	"at least one heads"
HT	{ "at least one heads" w.p. 0.5 "at least one tails" w.p. 0.5
TH	
TT	"at least one tails"

Let $A = \{ \text{info. provided is "at least one heads"} \}$
 $B = \{ \text{both flips are heads} \}$

$$P(B|A) = \frac{P(B)}{P(A)}$$

$$\begin{aligned}
 P(A) &= P(A|HH)P(HH) + P(A|HT)P(HT) \\
 &\quad + P(A|TH)P(TH) + P(A|TT)P(TT) \\
 &= (1)\left(\frac{1}{4}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{4}\right) + (0)\left(\frac{1}{4}\right) \\
 &= \frac{1}{2}, \text{ so } P(B|A) = \frac{1}{2}
 \end{aligned}$$

Version Three

<u>Result</u>	<u>Info.</u>
HH	Notice heads
HT	{ Notice heads w.p. 0.5 Notice tails w.p. 0.5
TH	
TT	Notice tails

$$A = \{\text{notice heads}\}$$

$$B = \{\text{both flips are heads}\}$$

$$P(B|A) = \frac{P(B)}{P(A)} = \frac{1/4}{1/2} = 1/2$$

H|H HT
TH ~~TT~~

Exercise (Based on HOTS): Consider the "broken-stick-into-a-triangle" question. What is the distribution of the length of the longest of the three pieces (regardless of whether or not a triangle can be formed)?

$U_1, U_2 \sim \text{uniform}(0,1)$ positions of cuts

U_1, U_2 independent

L = Length of longest piece. What is F_L ?

$$F_L(l) = \begin{cases} 0, & l < 1/3 \\ ?, & 1/3 \leq l \leq 1 \\ 1, & l > 1 \end{cases}$$

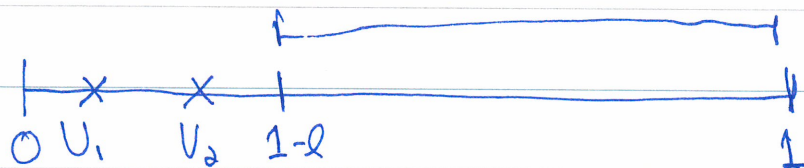
Fix $1/3 \leq l \leq 1$. When is $L \leq l$? When is $L > l$?

①



both $U_1, U_2 > l$

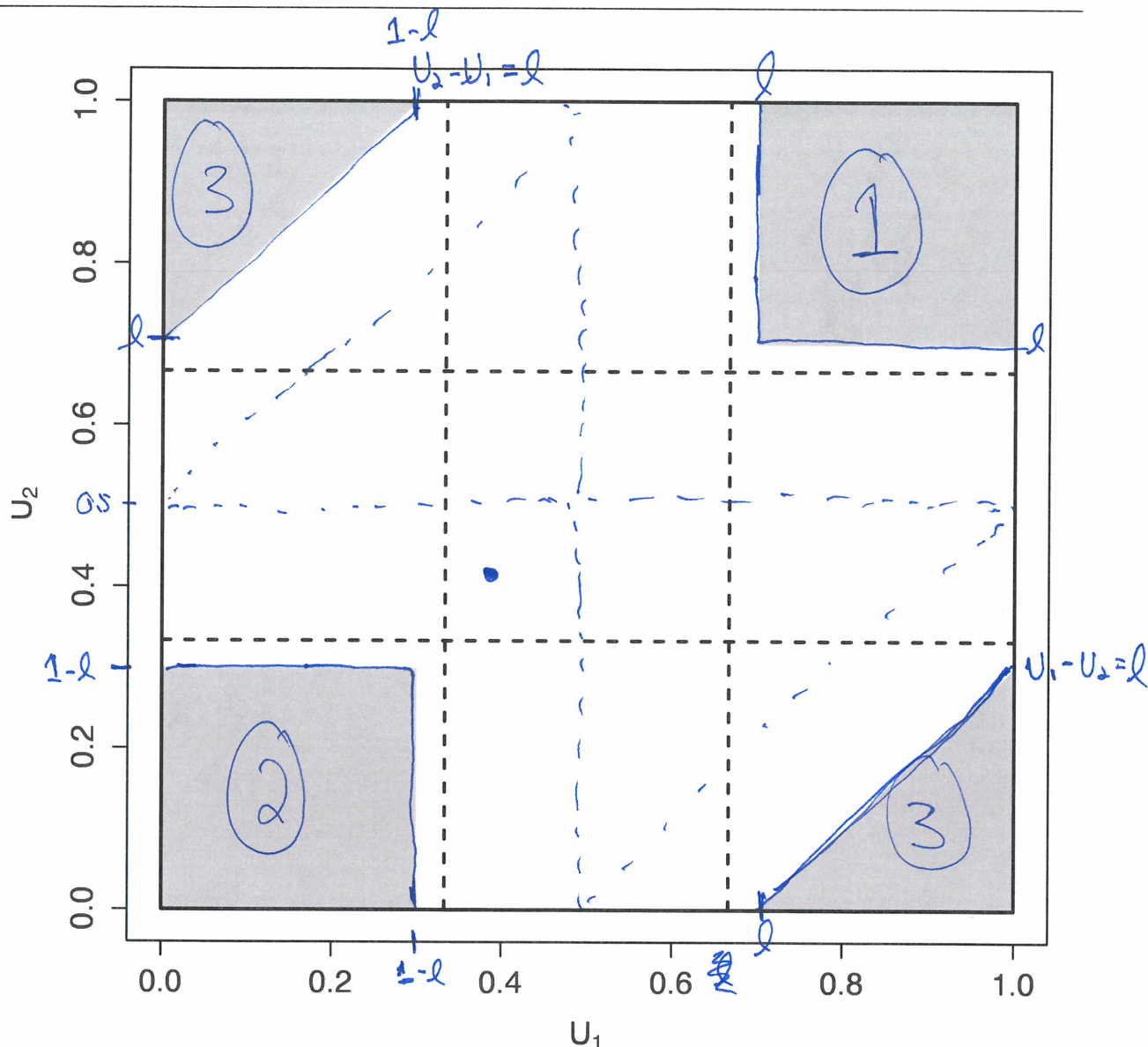
②



both $U_1, U_2 < 1-l$

③

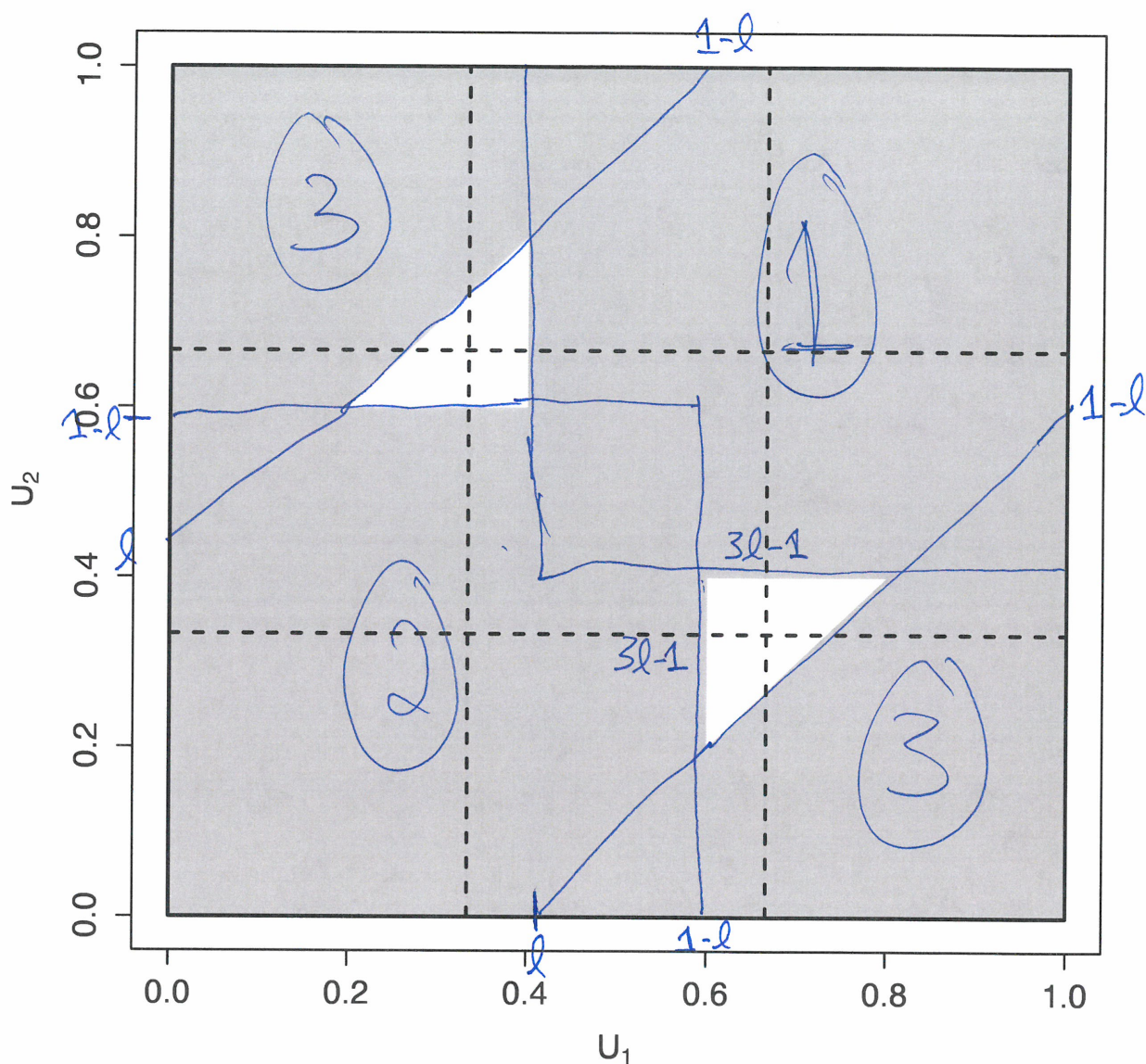




Case where $\frac{1}{2} \leq l \leq 1$

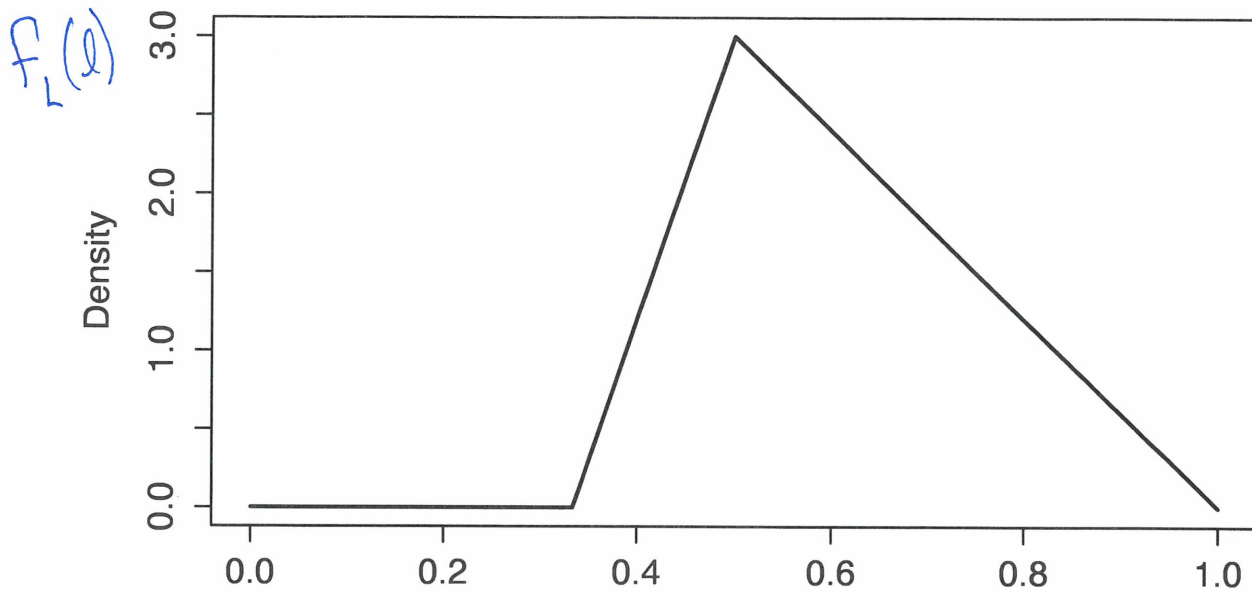
$$P(L > l) = \text{area of shaded region} \\ = 3(1-l)^2$$

$$P(L \leq l) = 1 - 3(1-l)^2$$



Case where $\frac{1}{3} \leq l < \frac{1}{2}$

$$P(L \leq l) = \text{area of unshaded region} \\ = (3l-1)^2$$



$$F_L(l) = \begin{cases} 0, & l < 1/3 \\ (3l-1)^2, & 1/3 \leq l < 1/2 \\ 1-3(1-l)^2, & 1/2 \leq l \leq 1 \\ 1, & l > 1 \end{cases}$$

$$f_L(l) = \begin{cases} 6(3l-1), & 1/3 \leq l \leq 1/2 \\ 6(1-l), & 1/2 \leq l \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$P(\text{form a } \Delta) = F_L(1/2) = 1/4$$

1 The Pareto(α, λ) Distribution

2 The random variable X has the **Pareto**(α, λ) **distribution** if $\alpha > 0$ and $\lambda > 0$
3 with

$$4 \quad f_X(x) = \begin{cases} \frac{\alpha \lambda^\alpha}{x^{\alpha+1}}, & \underline{x \geq \lambda} \\ 0, & \text{otherwise} \end{cases}.$$

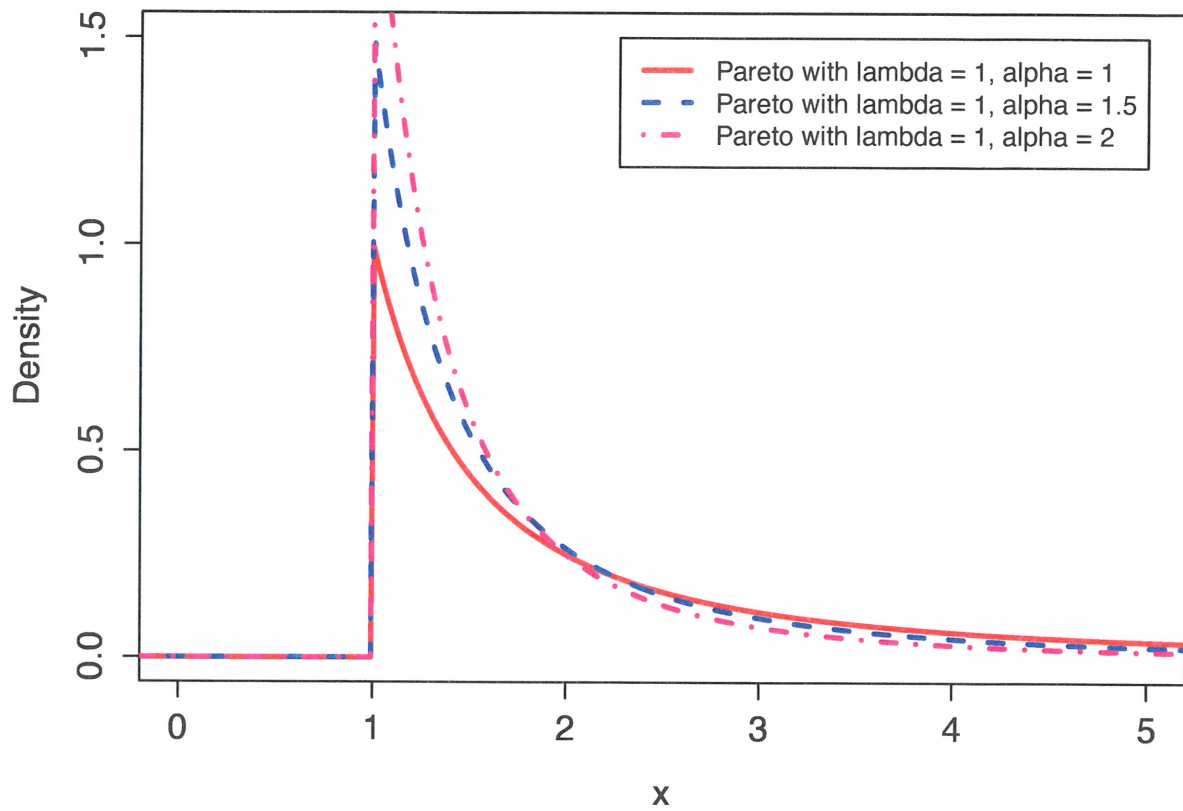
5 This is often used for modelling the tail of a distribution.

6 Some useful properties:

$$7 \quad E(X) = \begin{cases} \infty, & \alpha \leq 1 \\ \frac{\alpha \lambda}{\alpha - 1}, & \alpha > 1 \end{cases}$$

$$8 \quad V(X) = \begin{cases} \text{undefined}, & \alpha \leq 1 \\ \infty, & 1 < \alpha \leq 2 \\ \frac{\alpha \lambda}{(\alpha - 1)^2 (\alpha - 2)}, & \alpha > 2 \end{cases}$$

$$9 \quad F_X(x) = \begin{cases} 1 - \left(\frac{\lambda}{x}\right)^\alpha, & x \geq \lambda \\ 0, & x < \lambda \end{cases}$$



1

1 The Exponential(λ) Distribution

2 When X has the **Exponential(λ) distribution**, the following hold:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

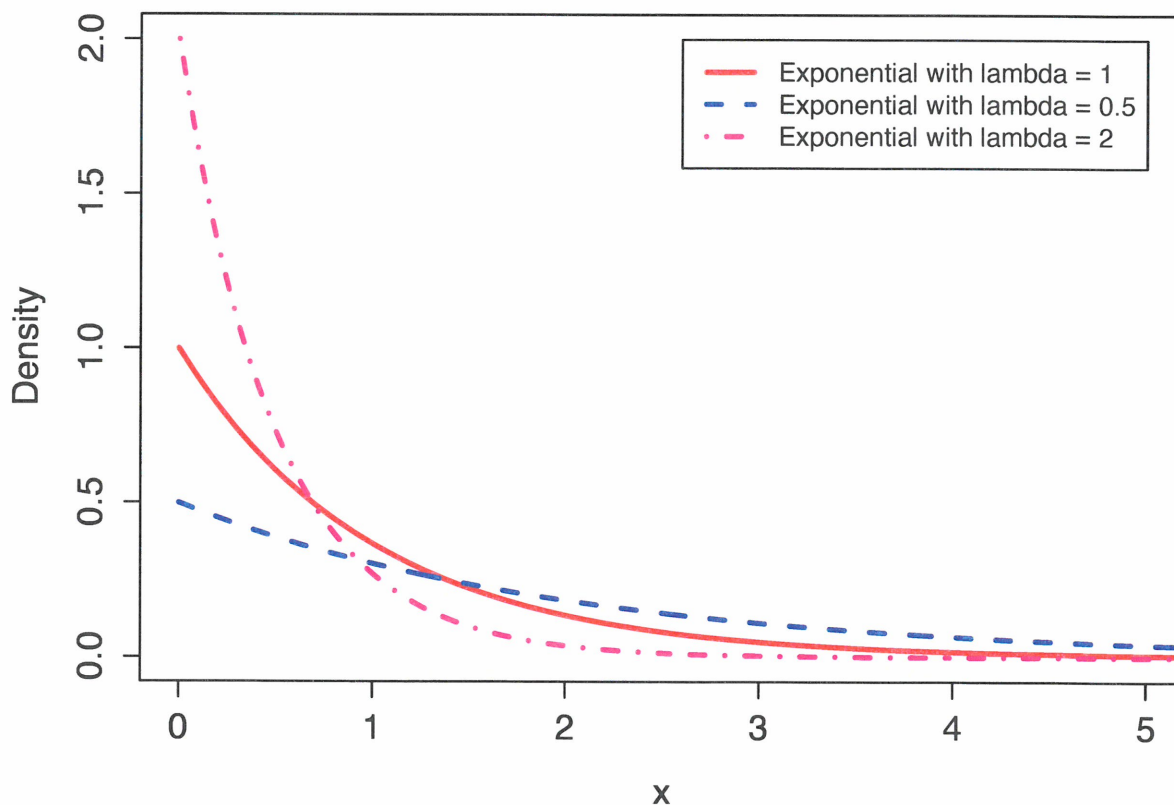
"rate parameterization"
 $\lambda = \text{rate}$

$$\underline{E(X) = 1/\lambda \text{ and } V(X) = 1/\lambda^2}$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$f_X(x) = \begin{cases} \frac{1}{\mathbb{E}} e^{-x/\mathbb{E}}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

"mean parameterization"
 $E(X) = \mathbb{E}$



Exercise: Suppose that X has the Exponential(λ) distribution, and that $a > 0$ and $b > 0$. Show that

$$P(X > a + b | X > b) = P(X > a).$$

Why is this called the "Memoryless Property?"

$$P(X > a + b | X > b) = \frac{P(X > a + b \text{ and } X > b)}{P(X > b)}$$

$$= \frac{P(X > a + b)}{P(X > b)}$$

$$= \frac{1 - F_X(a + b)}{1 - F_X(b)}$$

$$= \frac{e^{-\lambda(a+b)}}{e^{-\lambda b}}$$

$$= e^{-\lambda a}$$

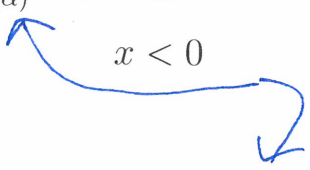
$$= P(X > a)$$

Point: If I model the times between "events" as exponential, the prob. the next event is at least a time units away does not depend on the time since the ~~last~~ previous event

"memoryless property"

1 The Gamma(α, β) Distribution

2 For $\alpha > 0$ and $\beta > 0$, random variable X has the **Gamma(α, β) distribu-**
3 **tion** when

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1} e^{-\beta x} \beta^\alpha}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$


5 where

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \text{ is the } \mathbf{\text{Gamma Function}}.$$

7 **Properties worth noting:** If $X \sim \text{Ga}(\alpha, \beta)$, then

8 1. $E(X) = \alpha/\beta$

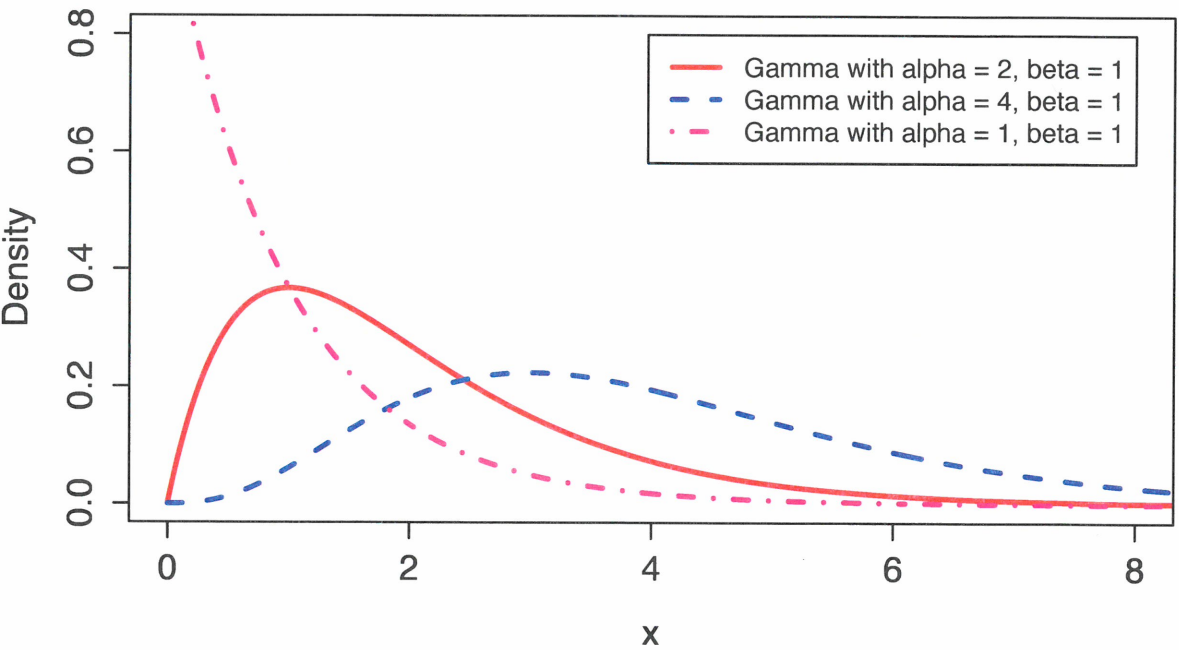
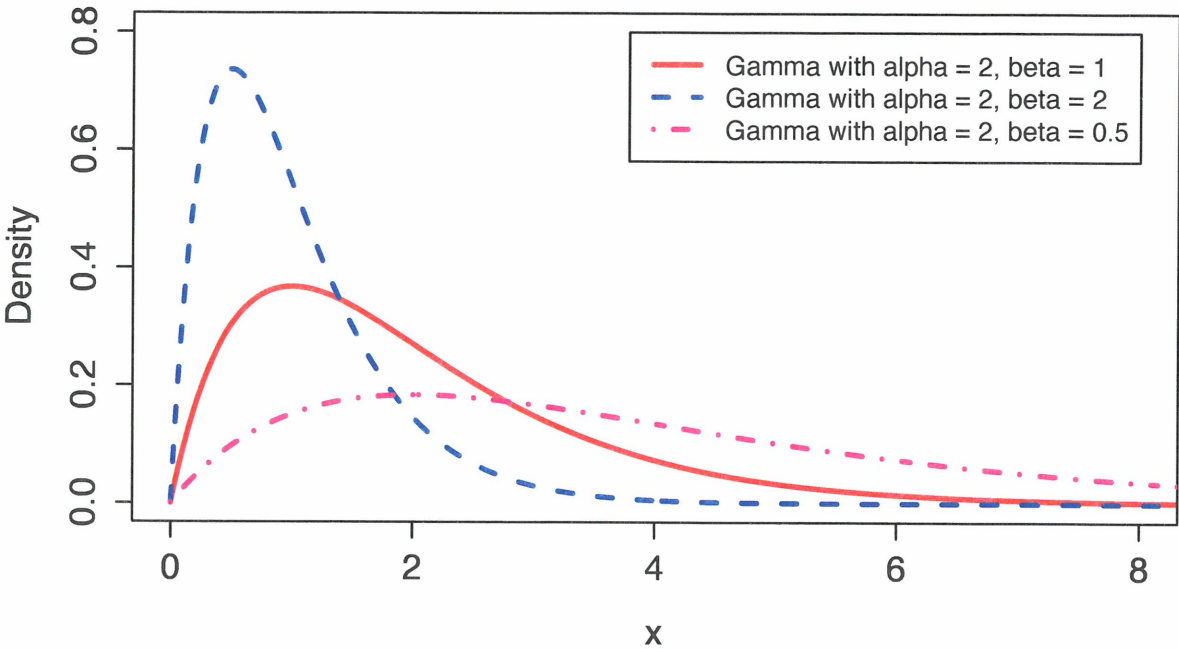
9 2. $V(X) = \alpha/\beta^2$

10 3. If $\alpha = 1$, then $X \sim \text{Exp}(\beta)$.

11 4. For any $c > 0$, the random variable $cX \sim \text{Ga}(\alpha, \beta/c)$.

12 5. If $X_1, X_2, \dots, X_n$ are independent random variables with $X_i \sim \text{Ga}(\alpha_i, \beta)$,
13 then $\sum_i X_i \sim \text{Ga}(\sum_i \alpha_i, \beta)$. 

14 6. The sum of n independent $\text{Exponential}(\beta)$ random variables has the
15 **Gamma(n, β) distribution. This is called the Erlang Distribution.**



1

Exercise: You are waiting in line at the express checkout of the grocery store. Three customers are waiting ahead of you and one customer is currently being served. The checkout time for an individual customer has an exponential distribution with mean of 20 seconds. What is the probability that you will still be waiting for service to begin two minutes from now?

Note that the exponential assumption allows me to calculate this prob. without knowing how long the person currently being served has already spent receiving service - memoryless prop.

$$T = \text{time until I start service (in minutes)} \\ = T_0 + T_1 + T_2 + T_3$$

T_0 = remaining service time for person receiving service

T_1 = service time for person 1 ahead of me

T_2 = " " " " 2 " " "

T_3 = " " " " 3 " " "

$$T_0, T_1, T_2, T_3 \sim \text{Exponential}(3) \text{ or } \text{Gamma}(1, 3)$$

If I assume T_0, T_1, T_2, T_3 independent then
 $T \sim \text{Gamma}(4, 3)$

1 The Normal(μ, σ^2) Distribution

2 The random variable X has the **Normal**(μ, σ^2) **Distribution** if, for $\sigma > 0$,

$$3 \quad f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad -\infty < x < \infty$$

4 This is sometimes called the **Gaussian Distribution**.

5 The case where $\mu = 0$ and $\sigma = 1$ is called the **standard normal distribution**.

6 The CDF of the standard normal distribution is commonly denoted with
7 Φ .

8 **Properties worth noting:** If $X \sim N(\mu, \sigma^2)$, then

9 1. $E(X) = \mu$

10 2. $V(X) = \sigma^2$

11 3. For $a \neq 0$ and any b , the random variable $aX + b$ has the $N(a\mu + b, a^2\sigma^2)$
12 distribution. Note that this implies that $(X - \mu)/\sigma$ has the standard
13 normal distribution.

$$a = 1/\sigma \quad b = -\mu/\sigma$$

14 4. Linear combinations of independent normal random variables are also
15 normally distributed.

Example: Suppose that $\log(S)$ has the $\text{Normal}(\mu, \sigma^2)$ distribution. Let K be a positive constant. Then it holds that

$$E((S - K)^+) = \exp(\mu + \sigma^2/2) \Phi\left(\frac{\mu + \sigma^2 - \log(K)}{\sigma}\right) - K \Phi\left(\frac{\mu - \log K}{\sigma}\right).$$

"S is lognormal(μ, σ^2)"

(This can be thought of as the expected value of a call option with strike price K , with S representing the price of the underlying stock on the maturity date.)

Proof: Denote $Y = \log(S)$. Hence, Y is $N(\mu, \sigma^2)$ and we can see that

$$\begin{aligned} E((S - K)^+) &= E((e^Y - K)^+) \\ &= \int_{-\infty}^{\infty} (e^y - K)^+ f_Y(y) dy \quad \text{by Rule for Lazy Statistician} \\ &= \int_{\log(K)}^{\infty} (e^y - K) f_Y(y) dy \quad \text{since } e^y < K \text{ when } y < \log(K) \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K \int_{\log(K)}^{\infty} f_Y(y) dy \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K P(Y > \log(K)) \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K P\left(Z > \frac{\log(K) - \mu}{\sigma}\right) \\ &\quad \text{where } Z \text{ has the standard normal distribution} \\ &= \int_{\log(K)}^{\infty} e^y f_Y(y) dy - K \Phi\left(\frac{\mu - \log(K)}{\sigma}\right) \\ &\quad \text{using the fact that } \Phi(x) = 1 - \Phi(-x) \end{aligned}$$

The remaining integral is tedious, but not difficult. It requires “completing the square” in the numerator of the expression that results when the normal density is plugged in for f_Y .

$$\begin{aligned}
 \int_{\log(K)}^{\infty} e^y f_Y(y) dy &= \int_{\log(K)}^{\infty} \frac{e^y}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{e^y}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2 - 2\mu y + \mu^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2 - 2\mu y + \mu^2 - 2\sigma^2 y}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2 - 2y(\mu + \sigma^2) + \mu^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \\
 &\quad \times \exp\left(-\frac{y^2 - 2y(\mu + \sigma^2) + (\mu + \sigma^2)^2 - (\mu + \sigma^2)^2 + \mu^2}{2\sigma^2}\right) dy \\
 &= \int_{\log(K)}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y - (\mu + \sigma^2))^2}{2\sigma^2}\right) \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) dy \\
 &= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) P(X > \log(K)) \\
 &\quad \text{where } X \sim N(\mu + \sigma^2, \sigma^2) \\
 &= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) P\left(Z > \frac{\log(K) - \mu - \sigma^2}{\sigma}\right) \\
 &\quad \text{where } Z \text{ is standard normal} \\
 &= \exp\left(\mu + \frac{\sigma^2}{2}\right) \Phi\left(\frac{\mu + \sigma^2 - \log(K)}{\sigma}\right)
 \end{aligned}$$

Substituting in this expression leads to the desired result. $\square$

The Poisson Process

In our initial discussion of the $\text{Poisson}(\lambda)$ distribution, we noted how the Poisson can be derived from the limit of $\text{binomial}(n, \lambda/n)$ random variables. There we imagined the interval $(0, 1)$ divided into n subintervals.



Now, imagine the same construction, but with the time interval extended to indefinitely. This leads to the **Poisson Process with rate λ** .

If I say that “events occur as a Poisson Process with rate λ ,” then

1. If Y equals the number of events during an interval of time of length T , then Y has the $\text{Poisson}(\lambda T)$ distribution.

2. The times between events are random variables with the $\text{Exponential}(\lambda)$ distribution. 

3. The times between events are independent random variables.

4. The numbers of events in disjoint time intervals are independent random variables.

5. The waiting time for k events has the $\text{Gamma}(k, \lambda)$ distribution. (This is also called the Erlang distribution.)

Exercise: Show that Property (1) above implies Property (2).

Suppose time 0 corresponds to an "event"

T = time of the subsequent event

For $t > 0$,

$$F_T(t) = P(T \leq t)$$

$$= P(\text{at least one event during } [0, t])$$

$$= 1 - P(\text{no events during } (0, t])$$

$$= 1 - \frac{e^{-\lambda t} (\lambda t)^0}{0!}$$

since # of events during $(0, t]$ has
Poisson (λt) distribution.

$$= 1 - e^{-\lambda t}$$

So, by uniqueness of CDF, conclude
that $T \sim \text{Exp}(\lambda)$

Exercise: Suppose that X_2 equals the number of “jumps” between now and maturity date for a call option. Assume that these jumps arrive as a Poisson process with rate λ (in units of day^{-1}). When a jump does occur, its size is normally distributed with mean zero and variance τ^2 . There is still an underlying normal distribution to model the “normal” fluctuations in the stock price on the log-scale. Let T denote the time to maturity (in days).

So, the model can be written as follows: If S is the price on the maturity date, then

$$S = \exp \left(X_1 + \sum_{i=1}^{X_2} Z_i \right)$$

where X_1 has the $\text{Normal}(\mu, \sigma^2)$ distribution, X_2 has the $\text{Poisson}(\lambda T)$ distribution, and the Z_i have the $\text{Normal}(0, \tau^2)$ distribution.

Let K be the strike price for the option. What is $E((S - K)^+)$? You can assume that all of the random variables are independent.

$$E((S - K)^+) = \sum_{j=0}^{\infty} E((S - K)^+ | X_2 = j) P(X_2 = j)$$

a version of LFTP for expectations
When $X_2 = j$, $X_1 + \sum_{i=1}^{X_2} Z_i \sim N(\mu, \sigma^2 + j\tau^2)$

Using the fact that U, V uncorrelated,
 $V(U + V) = V(U) + V(V)$
independence $\Rightarrow$ uncorrelated

So, $E((S-k)^+ | X_2=j)$ can be

found using the previous expression
(on page 50). Now, $\log(S)$ is conditioned on X_2

$N(\mu, \sigma^2 + j\tau^2)$ instead of

$N(\mu, \sigma^2)$

$$\text{Plus, } P(X_2=j) = \frac{e^{-\lambda T} (\lambda T)^j}{j!}$$

since $X_2 \sim \text{Poisson}(\lambda T)$

Properties of Expectation

If X, Y are random variables, and a, b , and c are constants, then

1. $E(c) = c$

2. $E(cX) = cE(X)$

3. $E(X + Y) = E(X) + E(Y)$ if either

(a) X, Y are both nonnegative

(b) X, Y are both nonpositive

(c) At least one of X, Y has finite expected values

4. $E(aX + b) = aE(X) + b$

5. $V(aX + b) = a^2V(X)$

$\infty - \infty$ $Y = -X$
 $E(X) = \infty$

Exercise: Suppose that a box contains n balls, labelled with numbers from 1 to n . Draws are made **with replacement** from the box m times, and the label on each ball is recorded after each draw. (Assume each ball is equally likely to be drawn.) What is the expected number of unique balls among the m draws?

$$X_i = \begin{cases} 1, & \text{if ball } i \text{ is drawn at least once} \\ 0, & \text{if not} \end{cases}$$

$i=1, 2, \dots, n$

$$\sum_{i=1}^n X_i = \# \text{ of unique balls}$$

$$E\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n E(X_i)$$

$$\begin{aligned} E(X_i) &= P(X_i = 1) = P(\text{ball } i \text{ drawn at least once}) \\ &= 1 - P(\text{ball } i \text{ never drawn}) = 1 - \left(\frac{n-1}{n}\right)^m \end{aligned}$$

$$E(\# \text{ of unique balls}) = n \left(1 - \left(\frac{n-1}{n}\right)^m\right)$$

Exercise: Two people play a game until one of the two players wins four rounds. Assume that the outcome of each round is independent, and each player is equally likely to win on each round. What is the expected number of times their score is tied? (Don't count 0 to 0.)

Can only tie at 1-1, 2-2, 3-3

$\{\text{tie at 1-1}\} = \{\text{players split first two games}\}$

$$\begin{aligned} \text{So, } P(\text{tie at 1-1}) &= P(\text{exactly one H in two flips}) \\ &= \binom{2}{1} 0.5^2 = \frac{1}{2} \end{aligned}$$

$$P(\text{tie at 2-2}) = \binom{4}{2} 0.5^4 = \frac{6}{16}$$

$$P(\text{tie at 3-3}) = \binom{6}{3} 0.5^6 = \frac{20}{64}$$

$$\# \text{ of ties} = 1_{\{\text{tie at 1-1}\}} + 1_{\{\text{tie at 2-2}\}} + 1_{\{\text{tie at 3-3}\}}$$

$$E(\# \text{ of ties}) = E(\quad) + E(\quad) + E(\quad)$$

$$= P(\text{tie at 1-1}) + P(\text{tie at 2-2}) + P(\text{tie at 3-3})$$

$$= \frac{76}{64}$$